INCH-POUND
MIL-DTL-24787/6A(SH)
24 September 2020
SUPERSEDING
MIL-F-24787/6(SH)
3 September 1993

DETAIL SPECIFICATION SHEET

FITTINGS, END, TYPE U (HOSE TO HOSE, 180-DEGREE RETURN), REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

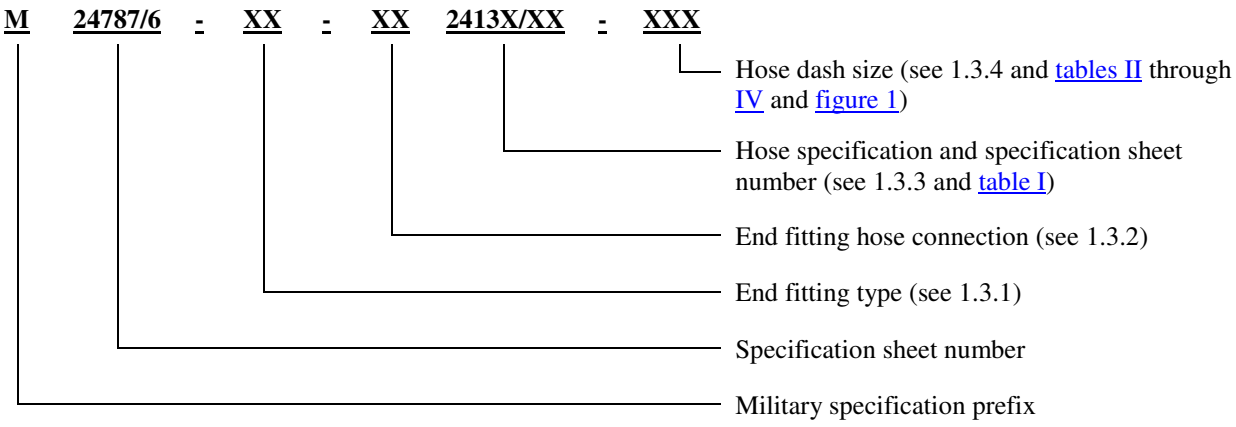
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24787.

1. SCOPE

1.1 Scope. This specification covers reusable hose to hose,180-degree return end fittings (type U) for hose sizes ranging from -24 to -192 for use with wire- or synthetic fiber-reinforced rubber flexible hose assemblies in accordance with MIL-DTL-24135 and MIL-DTL-24136.

1.2 Classification. The 180-degree return end fittings are type U as identified on figure 1. For the corresponding hose types, see table I.

1.3 Part or identifying number (PIN). The PIN is a definitive part number that is formed by combining the applicable specification sheet number (M24787/6 for MIL-DTL-24787/6), a dash, end fitting type, a dash, end fitting hose connection, hose specification and associated specification sheet number, a dash, and hose dash size.



PIN example: M24787/6-UB-RB24135/04-40X describes a MIL-DTL-24787/6 end fitting, type UB (U-bend), with a ring and band hose connection, for use on MIL-DTL-24135/4 and MIL-DTL-24135/10 (-40 and -48 only) hoses, with a hose size of -40.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

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1.3.1 End fitting type. End fitting types are identified in the PIN as follows:

End fitting type	PIN code
U-Bend	UB
Split Clamp	SC

1.3.2 End fitting hose connection. End fitting hose connections are identified in the PIN as follows:

Hose connection	PIN code
Ring and Band	RB
Bolt Together	BT

1.3.3 Hose specification and specification sheet number Specification sheet numbers that only contain one digit will enter "0" as the first digit. For example, for specification sheet 1, the PIN code will be recorded as "01". For end fittings categorized with multiple specification sheets, the first specification and applicable specification sheet will be used for the PIN. For example, for an end fitting that can be used with MIL-DTL-24135/4 and MIL-DTL-24135/10 (-40 and -48 only), the PIN code will be recorded as "24135/04".

1.3.4 Hose dash size. Hose dash sizes in tables II through IV only containing two digits will enter "X" as the last digit. For example, for a hose dash size of -40, the PIN code will be recorded as "40X". For hose dash sizes that contain three digits, all digits will be entered in the PIN. For example, for a hose dash size of -128, the PIN code will be recorded as "128".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24135 - Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specification for

MIL-DTL-24135/4 - Hose, Synthetic Rubber, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -40, -48, and -64)

MIL-DTL-24135/9 - Hose, Synthetic Rubber, Wire Reinforced (Four-Ply, Heavy Spiral-Wrapped), for Flexible Hose Assemblies (High Pressure, Sizes -8 through -32)

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MIL-DTL-24135/10	-	Hose, Synthetic Rubber, Wire Reinforced (Single- or Double-Wire Braid), for Flexible Hose Assemblies (Medium Pressure, Sizes -4 through -48)
MIL-DTL-24136	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24136/1	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies (Low Pressure, Sizes -80 through -192)
MIL-C-24723	-	Castings, Nickel-Copper Alloy
MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B271/B271M	-	Standard Specification for Copper-Base Alloy Centrifugal Castings
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM B505/B505M	-	Standard Specification for Copper Alloy Continuous Castings

(Copies these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. 180-degree return end fittings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 180-degree return end fittings. 180-degree return end fittings shall be in accordance with the requirements of MIL-DTL-24787 and as specified herein.

3.2.1 Dimensions. End fitting dimensions shall be in accordance with figure 1 and tables II through IV (see 6.2). All dimensions in the tables II through IV are in inches.

3.2.2 Materials. End fitting materials shall be in accordance with figure 1 and tables II through IV. Nickel-based investment castings, if used, shall be in accordance with the material requirements of ASTM A494/A494M, with the exception of M-25S, and the welding and non-destructive testing (NDT) requirements of MIL-C-24723. M-25S made in accordance with ASTM A494/A494M shall be in accordance with MIL-C-24723 for mechanical property, welding, and NDT requirements.

3.2.3 End fitting availability. 180-degree return end fittings (type U) are available in the hoses and sizes indicated in table I (see 6.2).

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TABLE I. 180-degree return end fitting by hose specification sheet and size.

End fitting size (hose dash number)	Hose specification sheet		
	24135/4 and /10 (-40 and -48)	24135/9	24136/1
-24	--	X	--
-40	X	--	--
-48	X	--	--
-64	X	--	--
-80	--	--	X
-96	--	--	X
-128	--	--	X
-160	--	--	X
-192	--	--	X
NOTE: X denotes where 180-degree return end fittings are available by specification sheet and size.			

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24787. Hose to hose, 180-degree return end fittings are not normally subjected to qualification testing since they do not attach to piping. They are qualified for a particular subgroup whenever an end fitting of that subgroup is submitted to and passes the requisite qualification testing.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24787.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. 180-degree return end fittings to this specification are intended for use in shipboard or shore-to-ship piping systems where pressures do not exceed the flange rated working pressure and temperatures do not exceed that of the applicable hose.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of end fittings.

c. Hose dash number (see 3.2.1).

d. End fitting type, class, hose specifications, and size (hose by pipe or tube) (see 3.2.3).

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24787 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

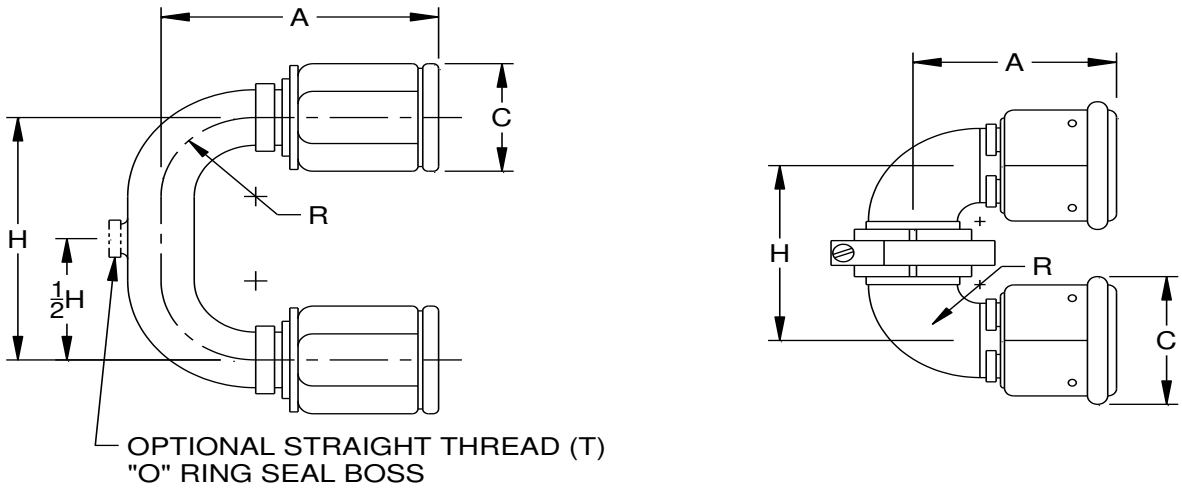
6.4 Subject term (key word) listing.

Synthetic fiber

Wire

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

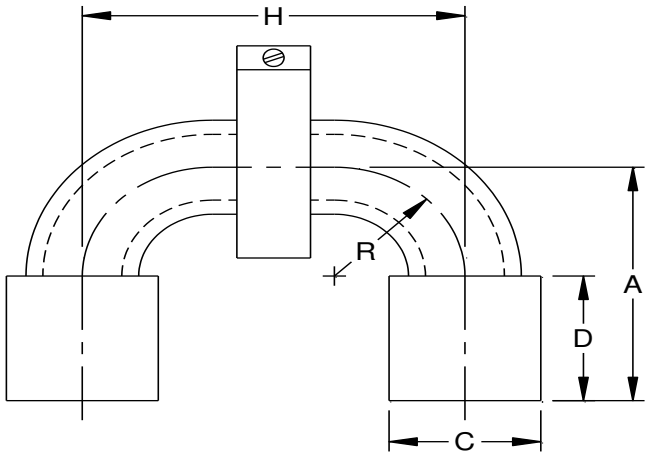
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U-BEND

SPLIT CLAMP

TYPE U - - - - - RING AND BAND



SPLIT CLAMP

TYPE U - - - - - BOLT TOGETHER

FIGURE 1. Type U end fittings.

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TABLE II. Dimensions for MIL-DTL-24135/9 hose to hose, 180-degree return with U-bend.

Hose size (dash number)	Boss thread UNF-2B	A max	C max	H nom	R min
-24	5/16 - 24	7.25	3.16	6.50	2.00

NOTES:

1. Sizes -20 and larger do not have this hex.
2. The end fitting assembly materials shall be in accordance with QQ-N-281 or QQ-N-286.
3. The socket alternative material shall be in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
4. The 90-degree tube elbow for sizes -24 and larger shall be in accordance with MIL-T-16420.

TABLE III. Dimensions for MIL-DTL-24135/4 and /10 (-40 and -48 only) hose to hose, 180-degree return with split clamp.

Hose size (dash number)	A max	C max	H nom	R min
-40	5.70	3.90	5.40	1.88
-48	6.05	4.50	5.92	2.09
-64	6.46	4.50	5.95	2.09

NOTES:

1. Sizes -20 and larger do not have this hex.
2. The end fitting assembly materials shall be in accordance with QQ-N-281 or QQ-N-286.
3. The socket alternative material shall be in accordance with MIL-C-24723 or ASTM A494/A494M. If ASTM A494/A494M is used, see 3.2.2 for additional guidance.
4. The 90-degree tube elbow for sizes -24 and larger shall be in accordance with MIL-T-16420.

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TABLE IV. Dimensions for MIL-DTL-24136/1 hose to hose, 180-degree return, bolt together.

Hose size (dash number)	A nom	C max	D nom	H max	R ref
-80	8.75	11.50	5.00	13.38	3.50
-96	9.25	12.00	5.00	14.14	4.00
-128	10.50	14.00	5.00	17.20	5.00
-160	12.00	16.50	5.00	19.20	6.38
-192	15.00	20.00	6.50	21.12	7.35

NOTES:

1. Material shall be UNS C95800 bronze in accordance with ASTM B148, ASTM B271/B271M, or ASTM B505/B505M.
2. Material for fasteners shall be in accordance with QQ-N-281 or QQ-N-286.

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CONCLUDING MATERIAL

Preparing activity:
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